

No. of Functional Features Included in the Solution

Date	27 June 2024
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Collects applicant personal, employment, and financial details through the web interface.	Frontend (Flask + HTML)	Done	High
2	Data Preprocessing	Handles missing values, encodes categorical data, and prepares data for prediction.	Data Processing	Done	High
3	Machine Learning Model Training	Trains Logistic Regression, Decision Tree, Random Forest, and XGBoost models.	ML Module	Done	High
4	Credit Card Approval Prediction	Predicts whether the application will be approved or rejected.	Prediction Engine	Done	High
5	Model Performance Evaluation	Compares models using Accuracy, Confusion Matrix, and Classification Report.	Evaluation Module	Done	Medium

6	Flask Web Application	Provides a user-friendly interface for prediction.	Web Application	Done	High
7	Prediction Result Display	Displays approval or rejection result instantly to the user.	Frontend	Done	Medium
8	Model Storage & Loading	Saves the best model using Pickle and loads it for real-time predictions.	Model Management	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Form, Prediction Result Page
2	Backend / Logic	4	Data Preprocessing, Model Training, Prediction, Evaluation
3	Database / Storage	1	Trained Model (model.pkl) and Dataset Storage
4	API / Integration	1	Flask Integration with Machine Learning Model
5	Security / Authentication	0 (or 1 if login is implemented)	Input Validation